Nawa — Architecture & Build Roadmap

From concept demo to a real platform where MENA founders build in public, earn credibility, and get funded.

Technical design document · Version 1.0 · September 2026
Stack: Angular · Supabase · OpenAI · Stripe (later) · Prepared for the Nawa founding team

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1 · Executive summary

Nawa today is a front-end-only Angular application. Every "backend" behaviour — accounts, posts, funding, AI, live streaming — is simulated in the browser using an in-memory store

persisted to `localStorage`. This document describes how to turn it into a real, multi-user platform on a startup budget, and sequences the work into six phases so value ships early and the hardest, most expensive, most regulated features come last.

The core decision is to build on **Supabase** (a managed Postgres backend providing database, authentication, file storage, realtime, and serverless functions) rather than assembling raw cloud primitives. This keeps fixed cost near zero while small, lets a one-to-two person team move fast, and — because Supabase is Postgres underneath — leaves a clean migration path to Azure or any cloud later, once traction justifies it.

All simulated behaviour is replaced with real implementations. The AI idea validator and roadmap generator call **OpenAI** for genuine, idea-specific analysis from day one — grounded with real market data (web-search sourced), never templated. Money (crowdfunding, marketplace, payouts) is deferred to Phase 5 because it carries both payment-provider constraints in Tunisia and a formal regulatory framework.

2 · Guiding principles & key decisions

DECISION	CHOICE	WHY
Backend	Supabase (managed Postgres)	Near-zero fixed cost, minimal ops, relational fit, no lock-in (plain Postgres underneath)
Keep the front end	Angular (existing)	UI is already built; we swap the mock store for real API calls
AI	Real OpenAI, server-side	No fake output. The API key lives in an Edge Function, never in the browser bundle
Market data	Sourced (web-grounded)	Founders may show analysis to investors; numbers must be defensible, not invented
Payments	Deferred to Phase 5	Stripe is unavailable in Tunisia; needs a regional gateway + crowdfunding licence
Sequencing	Credibility first, money last	Prove founders will build in public before investing in regulated money flows

> **Non-negotiable security rule**
>
> The OpenAI key (and later the Stripe secret key) must **never** appear in the Angular bundle. The browser ships to the public, so anything in it is public. All secret-bearing calls go through Supabase Edge Functions: `Angular → Edge Function → OpenAI/Stripe`.

3 · The target stack

Angular (SPA) Supabase Postgres Supabase Auth Supabase Storage Supabase Realtime Supabase Edge Functions OpenAI API Stripe / regional gateway (Phase 5) Mux / LiveKit (Phase 6)

CONCERN	SERVICE	NOTES
UI / client	Angular 20	Existing app; replace `Store` internals with a Supabase data service
Database	Supabase Postgres	Relational data + row-level security (RLS) as the primary authorization layer
Auth	Supabase Auth	Email + social login; issues JWTs the database trusts via RLS
Media	Supabase Storage	Avatars, post images, short video; served via CDN
Realtime	Supabase Realtime	DMs, chat rooms, notifications, live feed updates
Serverless logic	Edge Functions (Deno)	Anything needing a secret key or trusted compute: AI calls, payments, score recompute
AI	OpenAI (gpt-4o-mini)	Validator, roadmap, Copilot — called only from Edge Functions

4 · Target architecture

Three tiers: the Angular client in the browser, Supabase as the managed backend, and external services (OpenAI now; Stripe and a video provider later) reached only through Edge Functions.

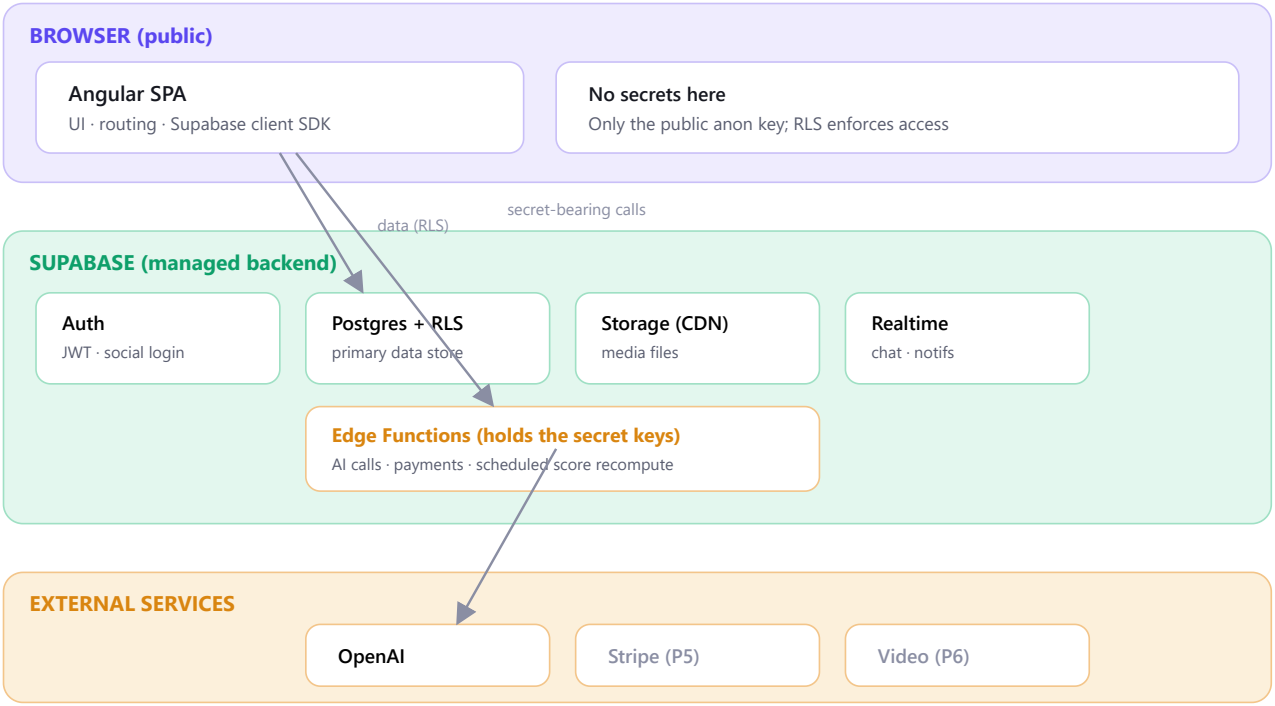


Figure 1 — Three-tier target architecture. The browser never holds a secret; Edge Functions mediate every call that needs one.

5 · Service inventory & Supabase coverage

Nawa is 18 services bundled into one app. The table shows how much each is covered by Supabase alone, and which phase delivers it.

Supabase covers fully Supabase + one external service Needs a specialist / regulated

SERVICE	COVERAGE	WHAT IT NEEDS BEYOND SUPABASE	PHASE
Identity & profiles	🟢	—	1
Startups (stages, KPIs, journey)	🟢	—	1
Build-in-public feed & follows	🟢	Storage for media	1
Founder Score	🟢	Edge Function (scheduled compute)	1
AI idea validator + roadmap	🟠	OpenAI via Edge Function	1
Discovery / search	🟢	Postgres full-text search	1
Direct messages	🟢	Realtime	2
Chat rooms	🟢	Realtime	2
Notifications	🟢	Realtime	2
Challenges & leaderboards	🟢	—	2
Stories / short video	🟠	Storage + CDN (transcoding later)	2
Copilot chat assistant	🟠	OpenAI via Edge Function	3
Experts (profiles, endorsements)	🟢	—	4
Incubators & cohorts	🟢	—	4
Jobs & events	🟢	—	4
Marketplace (orders)	🔴	Payment gateway	5
Donations / gifts	🔴	Payment gateway	5
Community rounds (crowdfunding)	🔴	Gateway + crowdfunding licence	5
Wallet / payouts	🔴	Connect-style payouts + KYC	5
Live streaming	🔴	Mux / LiveKit / Agora	6

Takeaway: Supabase covers roughly 70% of Nawa outright. The exceptions are AI (one external provider, proxied), money (a payment gateway plus regulation), and live video (a specialist).

6 · The phased roadmap

Phases are sequenced so the thesis — *founders build in public to earn credibility, then get funded* — is validated before any money or heavy infrastructure is built. Each phase ships something usable on its own.

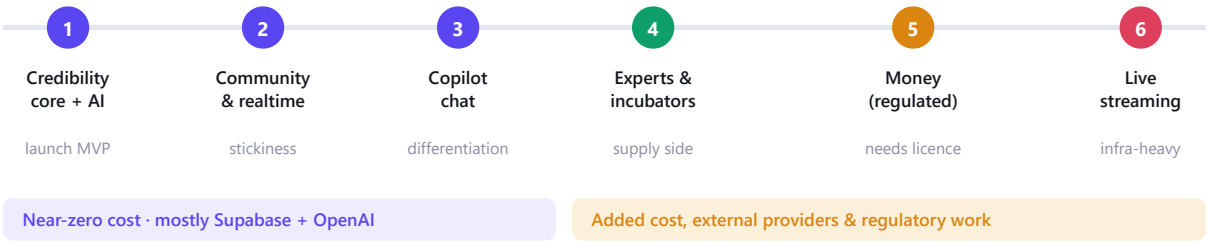


Figure 2 — Six phases. The first three are cheap and fast; cost and risk rise toward the right.

7 · Phase 1 — Credibility core + real AI onboarding MVP

Goal: a real founder signs up, describes their idea, gets a real AI validation and roadmap, creates a startup page, and builds in public with real followers and a real Founder Score. This is the smallest version that is genuinely useful and tests the core thesis.

Scope

- Authentication (email + social) and user profiles with roles
- Startup pages: stages, KPIs, journey, team
- Build-in-public feed: the eight post types, comments, supports, follows
- Real AI onboarding: idea validator + roadmap generator (see §8)
- Founder Score computed from real activity
- Discovery and search

Explicitly out of scope for Phase 1: messaging, live, experts, marketplace, funding — these are hidden or disabled until their phase.

Phase 1 data model (core tables)

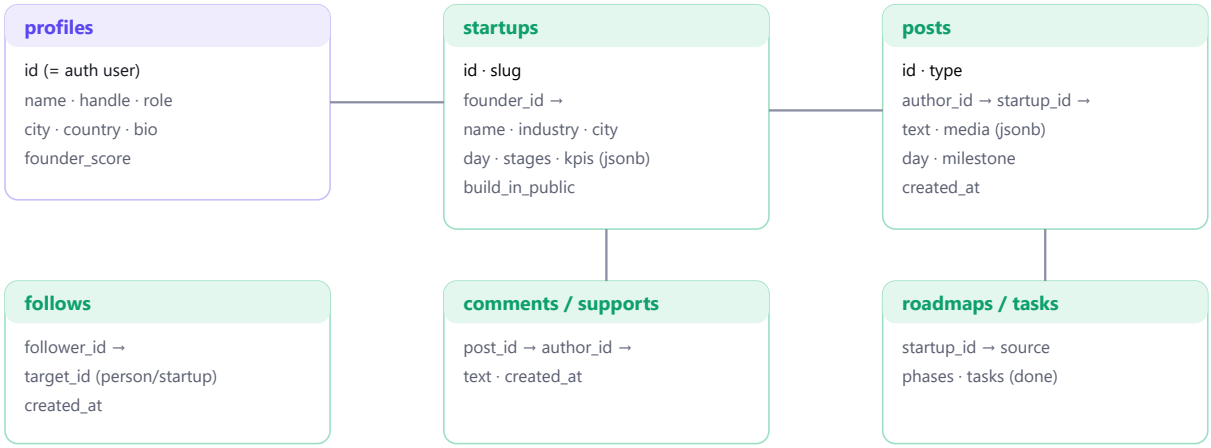


Figure 3 — Phase 1 relational model. Rich sub-structures (stages, KPIs, media) stay as JSONB; relationships that need querying (follows, comments) are their own tables.

Authorization via Row-Level Security (RLS)

Instead of a hand-written server, most access rules live in the database as RLS policies. Examples:

- Anyone can read public profiles, startups, and posts.
- A user can insert/update/delete a startup only where `founder_id = auth.uid()`.
- A user can comment or support as themselves only.

Migration plan — swap the mock Store for Supabase

The Angular `Store` stays as the app's single data access point; its internals change from `localStorage` arrays to Supabase queries. This is done table by table so the app keeps working throughout:

1. Create the Supabase project and apply the Phase 1 schema + RLS.
2. Seed reference data; wire Supabase Auth into the existing onboarding screens.
3. Replace `Store` read methods (profiles → startups → posts → follows) with Supabase reads.
4. Replace write methods (publish, follow, comment, support) with Supabase writes.
5. Move the Founder Score calculation into an Edge Function triggered on activity.
6. Remove `mock-data.ts` and the deterministic `ai.ts`; delete the admin "reset demo" control.

> ### Phase 1 exit criteria
>
> Real users can register, onboard with real AI analysis, publish updates, follow each other, and see a Founder Score that reflects real activity — with zero simulated data anywhere in the flow.

8 · The AI idea validator — prompt & contract

The validator is the first magical moment of the product, so it ships real in Phase 1. The flow:

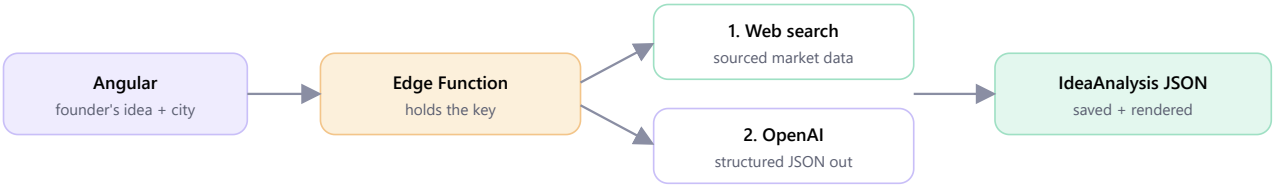


Figure 4 — Validator flow. The Edge Function grounds the analysis with sourced data (Option C), then asks the model for a strict JSON result.

Design choices

- **Model:** `gpt-4o-mini` — strong enough for structured analysis, ~16× cheaper than the full model.
- **Output:** OpenAI structured outputs (JSON schema mode) forcing a response that maps exactly to the existing `IdeaAnalysis` type — so it renders in the current UI with no reformatting.
- **Grounding (Option C):** the function first fetches real market indicators (search), then passes them to the model, which must cite them. Numbers it cannot ground are labelled as estimates.
- **Persona:** a candid MENA-focused startup analyst — specific, willing to push back on weak ideas, quantifying in local currency.

System prompt (draft — to be tuned by the team)

```
// system
You are a startup analyst for the Tunisian and wider MENA market, embedded in
Nawa, a build-in-public platform. You evaluate a founder's idea with honesty and
specificity. You are supportive but never flattering: if an idea is weak, you say
so and explain why.

Rules:
- Ground every market figure in the SOURCED DATA provided to you. If a number is
  not supported by that data, either omit it or clearly mark it "estimate".
- Quantify in the founder's local currency when known (TND for Tunisia).
- Name a concrete beachhead customer and a concrete next experiment.
- Prefer the realistic competitor (the status quo, a WhatsApp group, an
  intermediary) over naming big foreign startups.
- Never invent statistics, company names, or regulations.
- Return ONLY valid JSON matching the provided schema. No prose outside the JSON.
```

User prompt template

```
// user
FOUNDER IDEA:
"""{{idea}}"""

CONTEXT:
- City: {{city}}          - Country: {{country}}
- Stated role: {{role}}

SOURCED DATA (use for all market figures; cite in the notes fields):
{{search_results_json}}

Produce the full idea analysis as JSON matching the schema.
```

Enforced output schema (maps to `IdeaAnalysis`)

```
{
  "name": string,              // suggested startup name
  "tagline": string,
  "industry": enum,           // FinTech | AgriTech | AI | ...
  "emoji": string,
  "opportunityScore": number, // 0-100, justified
  "marketPotential": enum,    // Low | Medium | High
  "competition": enum,
  "risk": enum,
  "timing": string,
  "marketSize": [{ label, value, note }],  // note cites the source
  "competitors": [{ name, note, kind }],
  "strengths": string[],
  "risks": string[],
  "nextStep": string,
  "mvp": [{ label, why }],
  "pricing": string,
  "segments": string[]
}
```

Why Option C (sourced data) matters here

Founders will screenshot this analysis and show it to investors and incubators. A model left to invent TAM/SAM/SOM produces confident but unfounded numbers. Grounding the figures in real sources — and labelling anything unsourced as an estimate — protects both the founder's and Nawa's credibility.

9 · Phase 2 — Community & engagement

Goal: make the platform sticky and alive so founders return daily.

FEATURE	HOW IT IS BUILT
Direct messages (1:1)	Messages table + Supabase Realtime subscriptions; read receipts via a per-message flag
Chat rooms (startup / cohort / topic)	Rooms + membership tables; Realtime broadcast per room
Notifications	Notifications table; Realtime push to the recipient; Edge Function fan-out on events
Challenges & leaderboards	Relational; rankings computed by SQL queries over activity
Stories / short video	Supabase Storage + CDN; client-side upload with size limits

> **Exit criteria:** two founders can message in real time, join a room, receive live notifications, and compete on a challenge leaderboard.

10 · Phase 3 — The Copilot assistant

Goal: the differentiating "co-founder in your pocket" — ongoing, context-aware Q&A.

Unlike the one-shot validator, the Copilot answers open-ended questions ("why am I not getting customers?") using the founder's *real* startup state as context: current stage, KPIs, roadmap progress, follower count, campaign status. It runs through the same Edge Function pattern, injecting a compact context summary into each prompt. This is placed after community features because it is the clearest ongoing per-message cost and benefits from having real activity to reason about.

- Context builder: an Edge Function assembles a short, structured snapshot of the founder's startup.
- Guardrails: the assistant stays on startup topics and refuses to fabricate numbers about the user's own data.
- Cost control: short context, small model, optional per-user daily message cap.

11 · Phase 4 — Experts & incubators

Goal: bring in the supply side — mentorship and programmes.

FEATURE	PHASE 4 SCOPE	DEFERRED TO PHASE 5
Expert profiles	Profiles, categories, public answers, endorsements on founder profiles	—
Session booking	Free / calendar-based booking and scheduling	Paid sessions (needs payments)
Incubator portals	Cohorts, mentor assignment, custom roadmaps, cohort health dashboard	—
Jobs & events	Talent board, applications, demo days, AMAs, RSVPs	—

Endorsements feed back into the Founder Score, closing the credibility loop from Phase 1.

12 · Phase 5 — Money regulated

Goal: close the loop with real money — only after traction and legal footing. This is the hardest phase and has two distinct hard parts: a payment provider, and regulation.

Payment provider — Tunisia reality

Stripe is **not available** to Tunisia-based businesses; Tunisian founders typically only access it via a foreign (e.g. US) entity. For a Tunisia-first platform, plan around a local gateway authorised by the Banque Centrale de Tunisie — e.g. **Flouci** or **Konnect**, which accept national bank cards and mobile wallets in TND. Design the payments layer behind an interface so the specific gateway can be swapped per market.

Regulation — crowdfunding is a licensed activity

Tunisia regulates crowdfunding under **Law No. 2020-37 (6 August 2020)**, with application decrees published in October 2022 covering three distinct forms: investment in securities, lending, and donations/gifts. The securities-investment form is overseen by the CMF (financial market council). The practical implication for Nawa:

- Running a crowdfunding platform that collects money from the public is a *regulated activity*, not just a payments integration.
- Which form Nawa operates (rewards/donations vs. equity/lending) determines the licensing path and the regulator involved.
- This requires legal counsel before launch — it is a business/legal decision, not only an engineering one.

Build order within Phase 5 (increasing complexity)

1. **Marketplace payments** — simplest: a buyer pays a seller for a product via the gateway.
2. **Donations / tip jar** — direct support, same payment rails.
3. **Community rounds (crowdfunding)** — the regulated piece; gated on the legal path above.
4. **Wallet & payouts** — paying founders out, which adds KYC/identity obligations.

> **Recommendation:** begin the legal/licensing investigation at the *start* of the build (it has a long lead time) even though the code lands in Phase 5. Launch marketplace + donations first; open community rounds only once the regulatory path is cleared.

13 · Phase 6 — Live streaming

Goal: the richest, most infrastructure-heavy feature, built last.

Supabase has no video product, so live streaming uses a specialist provider — **Mux**, **LiveKit**, or **Agora**. Live chat and reactions run on Supabase Realtime; live gifts depend on the Phase 5 payment rails. This phase is last because it combines the highest ongoing infrastructure cost with a dependency on payments already being live.

14 · Cost model

All figures are current list prices from the sources in §16, in USD. They are indicative, not quotes.

PHASE / SCALE	SUPABASE	OPENAI (GPT-4O-MINI)	OTHER	INDICATIVE MONTHLY
Phases 1–2, pre-launch / small	Free tier ($0)	Pennies per analysis*	—	**≈ $0–10**
Phases 1–3, early traction	Pro ($25 base + usage)	Scales with signups & chats	Domain	**≈ $25–60**
Phase 5+, with payments	Pro + usage	Usage	Gateway % per txn	**$60+ + txn fees**
Phase 6, with live	Pro + usage	Usage	Video provider (per-minute)	**Variable, usage-led**

*gpt-4o-mini list price: $0.15 per 1M input tokens, $0.60 per 1M output tokens. A single idea analysis is a few thousand tokens — a fraction of a cent. Supabase Free is $0; Pro starts at $25/org/month plus usage-based overages. Supabase is billed per organisation with included quotas (database, storage, egress, monthly active users).

> The design keeps the only recurring per-user cost (AI calls) proportional to usage, and defers all fixed infrastructure cost until a phase genuinely needs it. Phases 1–2 can run at effectively zero fixed cost.

15 · Key risks & open questions

RISK / QUESTION	IMPACT	MITIGATION / DECISION NEEDED
Crowdfunding licensing (Law 2020-37)	High	Engage legal counsel early; decide rewards/donation vs. equity model; sequence money last
No native Stripe in Tunisia	Med	Integrate a BCT-authorised gateway (Flouci/Konnect) behind a swappable interface
AI market numbers must be defensible	Med	Option C: ground figures in sourced data; label estimates clearly
AI cost at scale	Low	Small model, short context, optional per-user caps
Arabic / Darija quality	Med	Evaluate model output on real dialect inputs; keep provider swappable
Live streaming infra cost	Med	Deferred to Phase 6; usage-based provider only when demand is proven

Open decisions needed from the team

1. Crowdfunding model: rewards/donations (lighter) or equity/lending (heavier regulation)?
2. Primary payment gateway for launch market (Flouci vs. Konnect vs. other)?
3. Social login providers to enable at launch (Google only, or more)?
4. Data residency: any requirement to host in a specific region?

16 · Sources

Market, pricing, and regulatory figures were drawn from public sources (accessed September 2026). Content was rephrased for compliance with licensing restrictions.

- OpenAI gpt-4o-mini pricing — openai.com, platform.openai.com/pricing
- Supabase pricing tiers — supabase.com/pricing and 2026 pricing analyses
- Stripe availability in Tunisia — stripe.com/global; Tunisian gateways Flouci / Konnect — flouci.com
- Tunisia crowdfunding framework — Law No. 2020-37 (6 Aug 2020) and application decrees (JORT, Oct 2022); CMF regulatory framework
- Tunisia startup ecosystem (1,450+ labelled startups, Startup Act / Law 2018-20) — Startup Tunisia (startup.gov.tn), Lucidity Insights 2025

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